

Diffusion and Reaction 15

Research is to see what everybody else sees, and
to think what nobody else has thought.

—Albert Szent-Gyorgyi

The concentration in
the internal surface
of the pellet is less
than that of the
external surface.

Overview. In this chapter we will develop models for diffusion with reaction in single-phase and two-phase systems. We start off with a discussion of diffusion and reaction in catalyst pellets. In Chapter 10, we assumed each point on the interior of the catalyst pellet's surface was accessible to the same concentration. However, when the reactants must diffuse inside the catalyst pellet's in order to react, we know the concentration at the pore mouth must be higher than that inside the pore. Consequently, the entire catalytic surface is not accessible to the same concentration; therefore, the rate of reaction throughout the pellet will vary. To account for variations in reaction rate throughout the pellet, we introduce a parameter known as the *effectiveness factor*, which is the ratio of the overall reaction rate in the pellet to the reaction rate at the external surface of the pellet.

The following topics will be discussed in this chapter:

- Diffusion and Reactions in Homogeneous Systems (Section 15.1).
- Diffusion and Reactions in Spherical Catalyst Pellets (Section 15.2).
- The Internal Effectiveness Factor (Section 15.3).
- Falsified Kinetics (Section 15.4).
- Overall Effectiveness Factor (Section 15.5).
- Estimation of Diffusion- and Reaction-Limited Regimes (Section 15.6).
- Mass Transfer and Reaction in a Packed Bed (Section 15.7).
- Determination of Limiting Situations from Reaction-Rate Data (Section 15.8).

After studying this chapter, you will be able to describe diffusion and reaction, determine when internal diffusion limits the overall rate of reaction for catalyst pellets, describe how to go about eliminating this limitation, and develop models for systems in which both diffusion and reaction play a role (e.g., catalysis, tissue growth).

15.1 Diffusion and Reactions in Homogeneous Systems

For homogeneous systems, the mole balance on species A, Equation (14-2), for one-dimensional diffusion at steady state is

$$-\frac{dW_{Az}}{dz} + r_A = 0$$

For diffusion through a stagnant film at dilute concentrations, Equation (14-9) becomes

$$W_{Az} = -D_{AB} \frac{dC_A}{dz}$$

Substituting in Equation (14-2) one obtains

$$D_{AB} \frac{d^2 C_A}{dz^2} + r_A = 0$$

Understanding and modeling diffusion with chemical reaction is not only important for industrial catalysts but also has many other applications. These applications include medicine, cancer treatment using drug laced particulates, and, as shown in the *Expanded Material* on the CRE Web site (www.umich.edu/~elements/5e/index.html), tissue engineering. In P15-15_B, we discuss the diffusion and reaction of oxygen in cartilage.

We will now discuss solid-gas catalytic reactions and diffusion limitation in catalyst pellets.

15.2 Diffusion and Reactions in Spherical Catalyst Pellets

The following sections of this chapter will focus solely on the transport and reaction in heterogeneous systems with catalyst pellets. In a heterogeneous reaction sequence, mass transfer of reactants must first take place from the bulk fluid to the external surface of the pellet. The reactants then diffuse from the external surface into and through the pores within the pellet ($C_{As} > C_A(r)$), with reaction taking place only on the catalytic surface of the pores. A schematic representation of this two-step diffusion process is shown in Figures 10-6, 14-1, and 15-1.

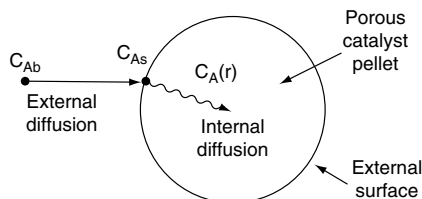


Figure 15-1 Mass transfer and reaction steps for a catalyst pellet.

In Chapter 14 we discussed *external diffusion*. In this section we will discuss *internal diffusion* and develop the internal effectiveness factor for spherical

catalyst pellets. The development of models that treat individual pores and pellets of different shapes is undertaken in the problems at the end of this chapter. We will first look at the internal mass transfer resistance to either the reaction products or reactants that occurs between the external pellet surface and the interior of the pellet. To illustrate the salient principles of this model, we consider the irreversible isomerization



that occurs on the surface of the pore walls within the spherical pellet of radius R .

15.2.1 Effective Diffusivity

To reach the inner surface, the reactant A must diffuse from a higher reactant concentration at the pellet external surface into and through the pores of pellets which are at a lower concentration as shown in Figure 15-1.

The pores in the pellet are not straight and cylindrical; rather, they are a series of tortuous, interconnecting paths of pore bodies and pore throats with varying cross-sectional areas. It would not be fruitful to describe diffusion within each and every one of the tortuous pathways individually; consequently, we shall define an effective diffusion coefficient so as to describe the average diffusion taking place at any interior position r in the pellet. We shall consider only radial variations in the concentration; the radial flux W_{Ar} will be based on the total area (voids and solid) normal to diffusion transport (i.e., $4\pi r^2$) rather than void area alone. This basis for W_{Ar} is made possible by proper definition of the effective diffusivity D_e .

The effective diffusivity accounts for the fact that:

1. Not all of the area normal to the direction of the flux is available (i.e., the area occupied by solids) for the molecules to diffuse.
2. The paths are tortuous.
3. The pores are of varying cross-sectional areas.

An equation that relates the effective diffusivity D_e to either the bulk diffusivity D_{AB} or the Knudsen diffusivity D_K is

$$D_e = \frac{D_{AB}\phi_p\sigma_c}{\tilde{\tau}} \quad (15-1)$$

The effective
diffusivity

where

$$\tilde{\tau} = \text{tortuosity}^1 = \frac{\text{Actual distance a molecule travels between two points}}{\text{Shortest distance between those two points}}$$

$$\phi_p = \text{pellet porosity} = \frac{\text{Volume of void space}}{\text{Total volume (voids and solids)}}$$

$$\sigma_c = \text{Constriction factor, see Figure 15-2(a)}$$

¹ Some investigators lump the constriction factor and tortuosity into one factor, called the tortuosity factor, and set it equal to $\tilde{\tau}/\sigma_c$. See C. N. Satterfield, *Mass Transfer in Heterogeneous Catalysis* (Cambridge, MA: MIT Press, 1970), pp. 33–47.

The constriction factor, σ_c , accounts for the variation in the cross-sectional area that is normal to diffusion.² It is a function of the ratio of maximum to minimum pore areas (Figure 15-2(a)). When the two areas, A_1 and A_2 , are equal, the constriction factor is unity, and when $\beta = 10$, the constriction factor is approximately 0.5.

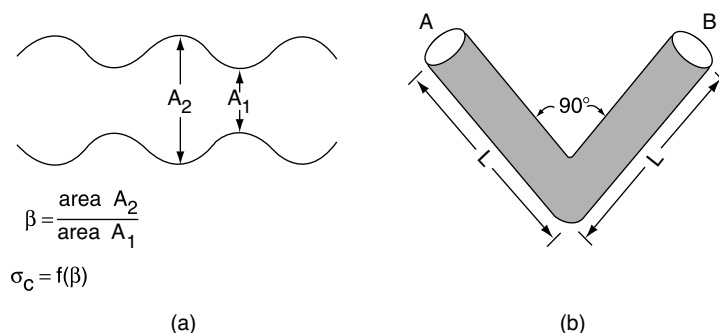


Figure 15-2 (a) Pore constriction; (b) pore tortuosity.

Example 15-1 Finding the Effective Diffusivity D_e

Using typical values of D_{AB} , ϕ_p , σ_c , and τ , estimate the effective diffusivity, D_e .

Solution

First, calculate the tortuosity for the hypothetical pore of length, L (Figure 15-2(b)), from the definition of $\tilde{\tau}$.

$$\tilde{\tau} = \frac{\text{Actual distance molecule travels from } A \text{ to } B}{\text{Shortest distance between } A \text{ and } B}$$

The shortest distance between points A and B for the idealized pore shown in Figure 15-2(b) is $\sqrt{2}L$. The actual distance the molecule travels from A to B is $2L$.

$$\tilde{\tau} = \frac{2L}{\sqrt{2}L} = \sqrt{2} = 1.414$$

Although this value is reasonable for $\tilde{\tau}$, values for $\tilde{\tau} = 6$ to 10 are not uncommon. Typical values of the constriction factor, the tortuosity, and the pellet porosity are, respectively, $\sigma_c = 0.8$, $\tilde{\tau} = 3.0$, and $\phi_p = 0.40$. A typical value of the gas=phase diffusivity is $D_{AB} = 10^{-6} \text{ m}^2/\text{s}$.

Using these values in Equation (15-1)

$$D_e = \frac{\phi_p \sigma_c}{\tilde{\tau}} D_{AB} \quad (15-1)$$

$$D_e = \frac{(0.4)0.8}{(3)} D_{AB} = 0.106 D_{AB}$$

² See E. E. Petersen, *Chemical Reaction Analysis* (Upper Saddle River, NJ: Prentice Hall, 1965), Chap. 3; C. N. Satterfield and T. K. Sherwood, *The Role of Diffusion in Catalysis* (Reading, MA: Addison-Wesley, 1963), Chap. 1.

therefore

$$D_e = 0.1 \cdot 10^{-6} \text{ m}^2/\text{s} = 10^{-7} \text{ m}^2/\text{s}$$

Analysis: The purpose of this example was to give an understanding of tortuosity and help the reader understand how it decreases D_e . We also see that a representative value of the effective diffusivity in the porous pellet is 10% of the gas-phase diffusivity.

15.2.2 Derivation of the Differential Equation Describing Diffusion and Reaction in a Single Catalyst Pellet

First, we will derive the concentration profile of reactant A in the pellet.

We now perform a steady-state mole balance on species A as it enters, leaves, and reacts in a spherical shell of inner radius r and outer radius $r + \Delta r$ of the pellet (Figure 15-3). Note that even though A is diffusing inward toward the center of the pellet, the convention of our shell balance dictates that the flux be in the direction of increasing r . We choose the flux of A to be positive in the direction of increasing r (i.e., the outward direction). Because A is actually diffusing inward, the flux of A will have some negative value, such as $(-10 \text{ mol}/\text{m}^2 \cdot \text{s})$, indicating that the flux is actually in the direction of decreasing r .

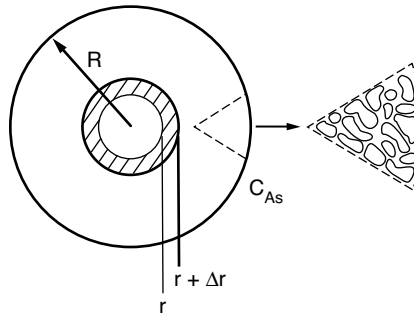


Figure 15-3 Shell balance on a catalyst pellet.

We now proceed to perform our shell balance on A. The area that appears in the balance equation is the total area (voids and solids) *normal* to the direction of the molar flux shown by the arrows in Figure 15-3.

$$\text{Rate of A in at } r = W_{Ar} \cdot \text{Area} = W_{Ar} \times 4\pi r^2 \Big|_r \quad (15-2)$$

$$\text{Rate of A out at } (r + \Delta r) = W_{Ar} \cdot \text{Area} = W_{Ar} \times 4\pi r^2 \Big|_{r+\Delta r} \quad (15-3)$$

$$\begin{aligned} \left[\begin{array}{c} \text{Rate of} \\ \text{generation} \\ \text{of A within a} \\ \text{shell of thickness} \\ \Delta r \end{array} \right] &= \left[\frac{\text{Rate of reaction}}{\text{Mass of catalyst}} \right] \times \left[\frac{\text{Mass catalyst}}{\text{Volume}} \right] \times [\text{Volume of shell}] \\ &= r'_A \times \rho_c \times 4\pi r_m^2 \Delta r \end{aligned} \quad (15-4)$$

Mole balance for diffusion and reaction inside the catalyst pellet

Mole balance

where r_m is some mean radius between r and $r + \Delta r$ that is used to approximate the volume ΔV of the shell, and ρ_c is the density of the pellet.

The mole balance over the shell thickness Δr is

$$\begin{aligned} &(\text{In at } r) - (\text{Out at } r + \Delta r) + (\text{Generation within } \Delta r) = 0 \\ &(W_{Ar} \times 4\pi r^2|_r) - (W_{Ar} \times 4\pi r^2|_{r+\Delta r}) + (r'_A \rho_c \times 4\pi r_m^2 \Delta r) = 0 \end{aligned} \quad (15-5)$$

After dividing by $(-4\pi \Delta r)$ and taking the limit as $\Delta r \rightarrow 0$, we obtain the following differential equation

$$\frac{d(W_{Ar} r^2)}{dr} - r'_A \rho_c r^2 = 0 \quad (15-6)$$

Because 1 mol of A reacts under conditions of constant temperature and pressure to form 1 mol of B, we have Equal Molar Counter Diffusion (EMCD) at constant total molar concentration (Section 14.2.1) and, therefore

The flux equation

$$W_{Ar} = -D_e \frac{dC_A}{dr} \quad (15-7)$$

where C_A is the number of moles of A per dm^3 of open-pore volume (i.e., volume of gas) as opposed to (mol/vol of gas and solids). In systems where we do not have EMCD in catalyst pores, it may still be possible to use Equation (15-7) if the reactant gases are present in dilute concentrations.

After substituting Equation (15-7) into Equation (15-6), we arrive at the following differential equation describing diffusion with reaction in a catalyst pellet

$$\boxed{\frac{d[-D_e(dC_A/dr)r^2]}{dr} - r^2 \rho_c r'_A = 0} \quad (15-8)$$

We now need to incorporate the rate law. In the past, we have based the rate of reaction in terms of either per unit volume

Inside the Pellet

$$\begin{aligned} -r'_A &= S_a (-r''_A) \\ -r_A &= \rho_c (-r'_A) \\ -r_A &= \rho_c S_a (-r''_A) \end{aligned}$$

$$\boxed{-r'_A [=] (\text{mol}/\text{dm}^3 \cdot \text{s})}$$

or per unit mass of catalyst

$$\boxed{-r'_A [=] (\text{mol}/\text{g-cat} \cdot \text{s})}$$

When we study reactions on the internal surface area of catalysts, the rate of reaction and rate law are often based on per unit surface area

$$\boxed{-r''_A [=] (\text{mol}/\text{m}^2 \cdot \text{s})}$$

As a result, the surface area of the catalyst per unit mass of catalyst

$$S_a [=] (\text{m}^2/\text{g-cat})$$

is an important property of the catalyst. The rate of reaction per unit mass of catalyst, $-r'_A$, and the rate of reaction per unit surface area of catalyst are related through the equation

$$-r'_A = -r''_A S_a$$



S_a : 10 grams of catalyst may cover as much surface area as a football field.

The rate law

A typical value of S_a might be $150 \text{ m}^2/\text{g}$ of catalyst.

As mentioned previously, at high temperatures, the denominator of the catalytic rate law often approaches 1 as discussed in Section 10.3.7. Consequently, for the moment, it is reasonable to assume that the surface reaction is of n th order in the gas-phase concentration of A within the pellet

$$-r_A'' = k_n'' C_A^n \quad (15-9)$$

where the units of the rate constants for $-r_A$, $-r_A'$, and $-r_A''$ are

$$-r_A'': \text{ with } k_n'' [=] \left(\frac{\text{m}^3}{\text{kmol}} \right)^{n-1} \frac{\text{m}}{\text{s}}$$

Similarly,

$$-r_A': \text{ with } k_n' = S_a k_n'' [=] \left(\frac{\text{m}^3}{\text{kmol}} \right)^{n-1} \frac{\text{m}^3}{\text{kg} \cdot \text{s}}$$

$$-r_A: \text{ with } k_n = k_n' \rho_c = \rho_c S_a k_n'' [=] \left(\frac{\text{m}^3}{\text{kmol}} \right)^{n-1} \frac{1}{\text{s}}$$

For a **first-order** catalytic reactor

per unit pellet surface area: $k_1'' = [\text{m/s}]$

per unit mass of single pellet catalyst: $k_1' = k_1'' S_a = [\text{m}^3/\text{kg} \cdot \text{s}]$

per unit single pellet volume: $k_1 = k_1' S_a \rho_c = [\text{s}^{-1}]$

Substituting the rate-law equation (15-9) into Equation (15-8) gives

$$\frac{d[r^2(-D_e dC_A/dr)]}{dr} + r^2 \overbrace{k_n'' S_a \rho_c}^{k_n} C_A^n = 0 \quad (15-10)$$

Differential equation and boundary conditions describing diffusion and reaction in a catalyst pellet

Letting k_n represent the terms under the bracket, differentiating the first term and dividing through by $-r^2 D_e$, Equation (15-10) becomes

$$\frac{d^2 C_A}{dr^2} + \frac{2}{r} \left(\frac{dC_A}{dr} \right) - \frac{k_n}{D_e} C_A^n = 0 \quad (15-11)$$

The boundary conditions are:

1. The concentration remains finite at the center of the pellet

$$C_A \text{ is finite at } r = 0$$

2. At the external surface of the catalyst pellet, the concentration is C_{As}

$$C_A = C_{As} \text{ at } r = R$$

15.2.3 Writing the Diffusion with the Catalytic Reaction Equation in Dimensionless Form

We now introduce dimensionless variables ψ and λ so that we may arrive at a parameter that is frequently discussed in catalytic reactions, the *Thiele modulus*. Let

$$\psi = \frac{C_A}{C_{As}} \quad (15-12)$$

$$\lambda = \frac{r}{R} \quad (15-13)$$

With the transformation of variables, the boundary condition

$$C_A = C_{As} \quad \text{at } r = R$$

becomes

$$\psi = \frac{C_A}{C_{As}} = 1 \quad \text{at } \lambda = 1$$

and the boundary condition

$$C_A \text{ is finite} \quad \text{at } r = 0$$

becomes

$$\psi \text{ is finite} \quad \text{at } \lambda = 0$$

We now rewrite the differential equation for the molar flux in terms of our dimensionless variables. Starting with

$$W_{Ar} = -D_e \frac{dC_A}{dr} \quad (15-7)$$

we use the chain rule to write

$$\frac{dC_A}{dr} = \left(\frac{dC_A}{d\lambda} \right) \frac{d\lambda}{dr} = \frac{d\psi}{d\lambda} \left(\frac{dC_A}{d\psi} \right) \frac{d\lambda}{dr} \quad (15-14)$$

Then differentiate Equation (15-12) with respect to ψ and Equation (15-13) with respect to r , and substitute the resulting expressions

$$\frac{dC_A}{d\psi} = C_{As} \quad \text{and} \quad \frac{d\lambda}{dr} = \frac{1}{R}$$

into Equation (15-14) to obtain

$$\frac{dC_A}{dr} = \frac{d\psi}{d\lambda} \frac{C_{As}}{R} \quad (15-15)$$

The flux of A in terms of the dimensionless variables, ψ and λ , is

The total rate of consumption of A inside the pellet, M_A (mol/s)

$$W_{Ar} = -D_e \frac{dC_A}{dr} = -\frac{D_e C_{As}}{R} \left(\frac{d\psi}{d\lambda} \right) \quad (15-16)$$

At steady state, the net flow of species A that enters into the pellet at the external pellet surface reacts completely within the pellet. The overall rate of reaction is therefore equal to the total molar flow of A into the catalyst pellet. The overall rate of reaction, M_A (mol/s), can be obtained by multiplying the molar flux into the pellet at the outer surface by the external surface area of the pellet, $4\pi R^2$

All the reactant that diffuses into the pellet is consumed (a black hole).

$$M_A = -4\pi R^2 W_{Ar} \Big|_{r=R} = +4\pi R^2 D_e \frac{dC_A}{dr} \Big|_{r=R} = 4\pi R D_e C_{As} \frac{d\psi}{d\lambda} \Big|_{\lambda=1} \quad (15-17)$$

Consequently, to determine the overall rate of reaction, which is given by Equation (15-17), we first solve Equation (15-11) for C_A , differentiate C_A with respect to r , and then substitute the resulting expression into Equation (15-17).

Differentiating the concentration gradient, Equation (15-15), yields

$$\frac{d^2 C_A}{dr^2} = \frac{d}{dr} \left(\frac{dC_A}{dr} \right) = \frac{d}{d\lambda} \left(\frac{d\psi}{d\lambda} \frac{C_{As}}{R} \right) \frac{d\lambda}{dr} = \frac{d^2 \psi}{d\lambda^2} \left(\frac{C_{As}}{R^2} \right) \quad (15-18)$$

After dividing by C_{As}/R^2 , the dimensionless form of Equation (15-11) is written as

$$\frac{d^2 \psi}{d\lambda^2} + \frac{2}{\lambda} \frac{d\psi}{d\lambda} - \frac{k_n R^2 C_{As}^{n-1}}{D_e} \psi^n = 0$$

Then

Dimensionless form of equations describing diffusion and reaction

$$\frac{d^2 \psi}{d\lambda^2} + \frac{2}{\lambda} \frac{d\psi}{d\lambda} - \phi_n^2 \psi^n = 0 \quad (15-19)$$

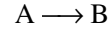
where

$$\phi_n^2 = \frac{k_n R^2 C_{As}^{n-1}}{D_e} \quad (15-20)$$

The square root of the coefficient of ψ^n in Equation 15-19 (i.e., ϕ_n) is called the Thiele modulus (pronounced *thē-lē*). The Thiele modulus, ϕ_n , will always contain a subscript (e.g., n), which refers to the reaction order and distinguishes this symbol from the symbol for porosity, ϕ , used in the Ergun pressure drop equation and defined in Chapter 5, which has no subscript. The quantity ϕ_n^2 is a measure of the ratio of “a” surface reaction rate to “a” rate of diffusion through the catalyst pellet

$$\phi_n^2 = \frac{k_n R^2 C_{As}^{n-1}}{D_e} = \frac{k_n R C_{As}^n}{D_e [(C_{As}-0)/R]} = \frac{\text{“a” surface reaction rate}}{\text{“a” diffusion rate}} \quad (15-20)$$

Limiting conditions: *When the Thiele modulus is large, internal diffusion usually limits the overall rate of reaction; when ϕ_n is small, the surface reaction is usually rate-limiting.* If, for the first order reaction



the reaction were surface reaction rate limited with respect to the adsorption of A and desorption of B, and if A and B were weakly adsorbed (i.e., low surface coverage), we can write the apparent first-order reaction rate law per unit volume of pellet as

$$-r_A = k_1 C_A$$

where k_1 is the rate constant for a single catalyst pellet.

Recalling $k_1 = S_a \rho_c k''$ we could also write the rate in terms of pellet catalytic surface area (mol/m²•s)

$$-r_A'' = k_1'' C_A \left(\frac{\text{mol}}{\text{m}^2 \cdot \text{s}} \right) \quad (15-21)$$

The units of k_1'' are m³/m²s (= m/s).

For a first-order reaction, Equation (15-19) becomes

$$\boxed{\frac{d^2 \psi}{d\lambda^2} + \frac{2}{\lambda} \frac{d\psi}{d\lambda} - \phi_1^2 \psi = 0} \quad (15-22)$$

the Thiele modulus for this first-order reaction is

$$\boxed{\phi_1 = R \sqrt{\frac{k_1'' \rho_c S_a}{D_e}} = R \sqrt{\frac{k_1}{D_e}}} \quad (15-23)$$

where

$$k_1 = k_1'' \rho_c S_a [=] \left(\frac{\text{m}}{\text{s}} \cdot \frac{\text{g}}{\text{m}^3} \cdot \frac{\text{m}^2}{\text{g}} \right) = 1/\text{s}$$

$$\frac{k_1}{D_e} [=] \left(\frac{1/\text{s}}{\text{m}^2/\text{s}} \right) = \frac{1}{\text{m}^2}$$

$$\phi_1 = R \sqrt{\frac{k_1}{D_e}} [=] \text{m} \left(\frac{\text{s}^{-1}}{\text{m}^2/\text{s}} \right)^{1/2} = \frac{1}{1} \quad (\text{Dimensionless})$$

The boundary conditions are

$$\boxed{\text{B.C. 1: } \psi = 1 \quad \text{at } \lambda = 1} \quad (15-24)$$

$$\boxed{\text{B.C. 2: } \psi \text{ is finite} \quad \text{at } \lambda = 0} \quad (15-25)$$

15.2.4 Solution to the Differential Equation for a First-Order Reaction

Differential equation (15-22) is readily solved with the aid of the transformation $y = \psi\lambda$:

$$\frac{d\psi}{d\lambda} = \frac{1}{\lambda} \left(\frac{dy}{d\lambda} \right) - \frac{y}{\lambda^2}$$

$$\frac{d^2\psi}{d\lambda^2} = \frac{1}{\lambda} \left(\frac{d^2y}{d\lambda^2} \right) - \frac{2}{\lambda^2} \left(\frac{dy}{d\lambda} \right) + \frac{2y}{\lambda^3}$$

With these transformations, Equation (15-22) reduces to

$$\frac{d^2\psi}{d\lambda^2} - \phi_1^2 \psi = 0 \quad (15-26)$$

This differential equation has the following solution (see Appendix A.3):

$$y = A_1 \cosh \phi_1 \lambda + B_1 \sinh \phi_1 \lambda$$

In terms of ψ ,

$$\psi = \frac{A_1}{\lambda} \cosh \phi_1 \lambda + \frac{B_1}{\lambda} \sinh \phi_1 \lambda$$

The arbitrary constants A_1 and B_1 can easily be evaluated with the aid of the boundary conditions. At $\lambda = 0$, $\cosh \phi_1 \lambda \rightarrow 1$, $(1/\lambda) \rightarrow \infty$, and $\sinh \phi_1 \lambda \rightarrow 0$. Because the second boundary condition requires ψ to be finite at the center (i.e., $\lambda = 0$), therefore A_1 must be zero.

The constant B_1 is evaluated from B.C. 1 (i.e., $\psi = 1$, $\lambda = 1$) and the dimensionless concentration profile is

Concentration
profile

$$\psi = \frac{C_A}{C_{As}} = \frac{1}{\lambda} \left(\frac{\sinh \phi_1 \lambda}{\sinh \phi_1} \right) \quad (15-27)$$

Figure 15-4 shows the concentration profile for three different values of the Thiele modulus, ϕ_1 . Small values of the Thiele modulus indicate surface reaction controls and a significant amount of the reactant diffuses well into the pellet interior without reacting. As a result, the concentration profile is very shallow, with the concentration at the center of the pellet being close to that at the external surface. That is, virtually the entire internal surface is accessible to the reactant concentration C_{As} . Large values of the Thiele modulus indicate that the surface reaction is rapid and that the reactant is consumed very close to the external pellet surface and very little penetrates into the interior of the pellet. Consequently, if the porous pellet is to be plated with a precious metal catalyst (e.g., Pt), it should only be plated in the immediate vicinity of the external surface when large values of ϕ_n characterize the diffusion and reaction. That is, it would be a waste of the precious metal to plate the entire pellet when internal diffusion is limiting because the reacting gases are consumed near the outer surface. Consequently, the reacting gases would never contact the center portion of the pellet.

For large values of the Thiele modulus, internal diffusion limits the rate of reaction.

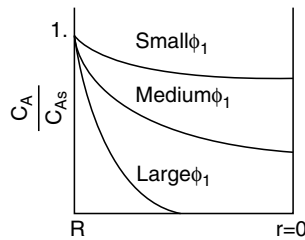


Figure 15-4 Concentration profile in a spherical catalyst pellet.

15.3 The Internal Effectiveness Factor

In Figure 15-4 we saw that the concentration varied with the pellet radius. Consequently, for all but zero-order reactions, the rate will also vary throughout the pellet. In order to account for this variation in rates, we introduce the *internal effectiveness factor*:

15.3.1 Isothermal First-Order Catalytic Reactions

The magnitude of the effectiveness factor (ranging from 0 to 1) indicates the relative importance of diffusion and reaction limitations. The internal effectiveness factor is defined as

η is a measure of how far the reactant diffuses into the pellet before reacting.

$$\eta = \frac{\text{Actual overall rate of reaction}}{\text{Rate of reaction that would result if entire interior surface exposed to the external pellet surface conditions } C_{As}, T_s} \quad (15-28)$$

The overall rate, $-r'_A$, is also referred to as the observed rate of reaction $[-r_A(\text{obs})]$. In terms of symbols, the effectiveness factor is

$$\eta = \frac{-r_A}{-r_{As}} = \frac{-r'_A}{-r'_{As}} = \frac{-r''_A}{-r''_{As}}$$

To derive the effectiveness factor for a first-order reaction, it is easiest to work in reaction rates of moles per unit time, M_A , rather than in moles per unit time per volume of catalyst (i.e., $-r_A$)

$$\eta = \frac{-r_A}{-r_{As}} = \frac{-r_A \times \text{Volume of catalyst particle}}{-r_{As} \times \text{Volume of catalyst particle}} = \frac{M_A}{M_{As}}$$

First, we shall consider the denominator, M_{As} . If the entire surface were exposed to the concentration at the external surface of the pellet, C_{As} , the rate for a first-order reaction would be

$$M_{As} = \frac{\text{Rate at external surface}}{\text{Volume}} \times \text{Volume of catalyst pellet}$$

$$M_{As} = -r_{As} \times \left(\frac{4}{3} \pi R^3 \right) = k_1 C_{As} \left(\frac{4}{3} \pi R^3 \right) \quad (15-29)$$

The subscript s indicates that the rate $-r_{As}$ is evaluated at the conditions (e.g., concentration, temperature) present at the **external** surface of the pellet (i.e., $\lambda = 1$).

The *actual rate of reaction* is the rate at which the reactant diffuses into the pellet at the outer surface; that is, all of A that diffuses into the pellet at the outer surface reacts and no A diffuses back out. (It behaves as a “black hole.”) We recall Equation (15-17) on page 727 for the actual rate of reaction

$$M_A = 4\pi R D_e C_{As} \left. \frac{d\psi}{d\lambda} \right|_{\lambda=1} \quad (15-17)$$

Differentiating Equation (15-27) and then evaluating the result at $\lambda = 1$ yields

$$\left. \frac{d\psi}{d\lambda} \right|_{\lambda=1} = \left(\frac{\phi_1 \cosh \lambda \phi_1}{\lambda \sinh \phi_1} - \frac{1}{\lambda^2} \frac{\sinh \lambda \phi_1}{\sinh \phi_1} \right)_{\lambda=1} = (\phi_1 \coth \phi_1 - 1) \quad (15-30)$$

Substituting Equation (15-30) into (15-17) gives us

$$M_A = 4\pi R D_e C_{As} (\phi_1 \coth \phi_1 - 1) \quad (15-31)$$

We now substitute Equations (15-29) and (15-31) into Equation (15-28) to obtain an expression for the effectiveness factor

$$\begin{aligned} \eta &= \frac{M_A}{M_{As}} = \frac{M_A}{(-r_{As}) \left(\frac{4}{3} \pi R^3 \right)} = \frac{4\pi R D_e C_{As}}{k_1 C_{As} \frac{4}{3} \pi R^3} (\phi_1 \coth \phi_1 - 1) \\ &= 3 \underbrace{\frac{1}{k_1 R^2 / D_e}}_{\phi_1^2} (\phi_1 \coth \phi_1 - 1) \end{aligned}$$

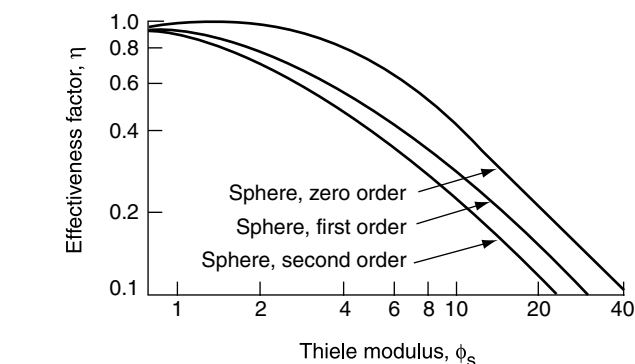
Internal
effectiveness factor
for a first-order
reaction in a
spherical catalyst
pellet

$$\eta = \frac{3}{\phi_1^2} (\phi_1 \coth \phi_1 - 1) \quad (15-32)$$

A plot of the effectiveness factor as a function of the Thiele modulus is shown in Figure 15-5. Figure 15-5(a) shows η as a function of the Thiele modulus, ϕ_s , for a spherical catalyst pellet for reactions of zero, first, and second order. Figure 15-5(b) corresponds to a first-order reaction occurring in three differently shaped pellets of volume V_p and external surface area A_p , and where the Thiele modulus for a first-order reaction, ϕ_1 , is defined differently for each shape. When volume change accompanies a reaction (i.e., $\varepsilon \neq 0$), the corrections shown in Figure 15-6 apply to the effectiveness factor for a first-order reaction.

We observe that as the particle diameter becomes very small, ϕ_n decreases, so that the effectiveness factor approaches 1 and the reaction is surface reaction-limited. On the other hand, when the Thiele modulus ϕ_n is large (e.g., ~ 30), the internal effectiveness factor η is small (e.g., ~ 0.1), and the reaction is

If $\phi_1 > 2$ then $\eta \approx \frac{3}{\phi_1^2} [\phi_1 - 1]$ If $\phi_1 > 20$ then $\eta \approx \frac{3}{\phi_1}$
--

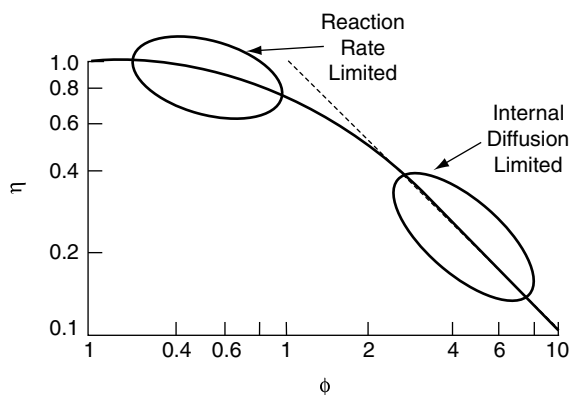


Zero order $\phi_{s0} = R\sqrt{k_0'' S_a \rho_c / D_e C_{A0}} = R\sqrt{k_0'' / D_e C_{A0}}$

First order $\phi_{s1} = R\sqrt{k_1'' S_a \rho_c / D_e} = R\sqrt{k_1'' / D_e}$

Second order $\phi_{s2} = R\sqrt{k_2'' S_a \rho_c C_{A0} / D_e} = R\sqrt{k_2 C_{A0} / D_e}$

(a)



Sphere $\phi_1 = (R/3)\sqrt{k_1'' S_a \rho_c / D_e} = \frac{R}{3}\sqrt{k_1 / D_e}$

Cylinder $\phi_1 = (R/2)\sqrt{k_1'' S_a \rho_c / D_e} = \frac{R}{2}\sqrt{k_1 / D_e}$

Slab $\phi_1 = L\sqrt{k_1'' S_a \rho_c / D_e} = L\sqrt{k_1 / D_e}$

(b)

Figure 15-5 (a) Effectiveness factor plot for n th-order kinetics on spherical catalyst particles. [Satterfield, C. N. "Mass Transfer in Heterogeneous Catalysis," *AIChE Journal* 16(3) 509–510 (1970). With permission of the American Institute of Chemical Engineers. Copyright © 1970 AIChE. All rights reserved.] (b) First-order reaction in different pellet geometries. [Aris, R. *Introduction to the Analysis of Chemical Reactors*, 1965, p. 131; reprinted by permission of Pearson Education, Upper Saddle River, NJ.]

diffusion-limited within the pellet. Consequently, factors influencing the rate of external mass transport such as fluid velocity will have a negligible effect on the overall reaction rate when the reaction is either internal surface reaction rate limited or internal diffusion limited. For large values of the Thiele modulus, the effectiveness factor can be written as

$$\eta \approx \frac{3}{\phi_1} = \frac{3}{R} \sqrt{\frac{D_e}{k_1}} \quad (15-33)$$

To express the overall rate of reaction in terms of the Thiele modulus, we rearrange Equation (15-28) and use the rate law for a first-order reaction in Equation (15-29)

$$\begin{aligned} -r_A &= \left(\frac{\text{Actual reaction rate}}{\text{Reaction rate at } C_{As}} \right) \times (\text{Reaction rate at } C_{As}) \\ &= \eta (-r_{As}) \end{aligned}$$

For a first-order reaction

$$-r_A = \eta (k_1 C_{As}) \quad (15-34)$$

Combining Equations (15-33) and (15-34), the overall rate of reaction for a first-order, internal-diffusion-limited reaction is

Note that for catalytic packed beds the rate varies inversely with the particle diameter of the catalyst.

$$-r_A = \frac{3}{R} \sqrt{D_e k_1} C_{As} = \frac{3}{R} \sqrt{D_e S_a \rho_c k_1''} C_{As}$$

15.3.2 Effectiveness Factors with Volume Change with Reaction

When there is volume change, $\varepsilon \neq 0$, we use a correction factor to account for this change. The correction is obtained from a plot of the ratio of effective factors

$$\frac{\eta'}{\eta} = \frac{\text{Factor in the presence of volume change}}{\text{Factor in the absence of volume change}}$$

as a function of ε for various values of the Thiele modulus. This plot is given in Figure 15-6.

For example, if the Thiele modulus were 10 for the gas-phase reaction $A \rightarrow 2B$ with ($\varepsilon = 1$) then the effectiveness factor with volume change would be $\eta' = 0.8\eta$.

15.3.3 Internal Diffusion Limited Reactions Other Than First Order

How can the rate of reaction be increased?

To increase the reaction rate, $-r'_A$, for internal diffusion limited reactions, we can (1) decrease the radius R (make pellets smaller); (2) increase the temperature; (3) increase the concentration; and (4) increase the internal surface area. For reactions of order n , we have from Equation (15-20)

$$\phi_n^2 = \frac{k_n'' S_a \rho_c R^2 C_{As}^{n-1}}{D_e} = \frac{k_n R^2 C_{As}^{n-1}}{D_e} \quad (15-20)$$